Decision Trees

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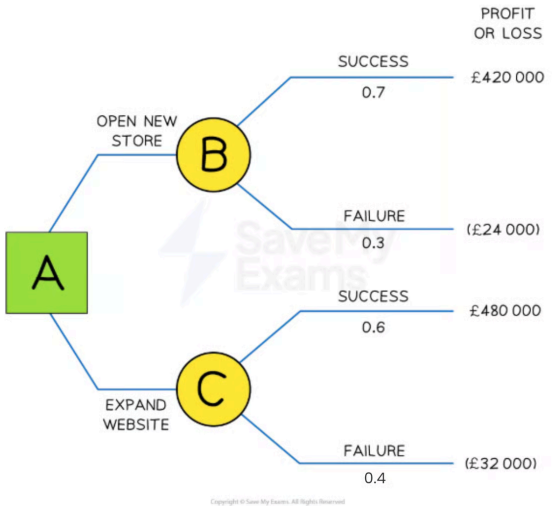
Decision tree diagrams

- A decision tree is a **quantitative method** of tracing the outcomes of a decision so that the most profitable decision can be identified
  - **Research-based estimates** and **probabilities** are used to calculate likely outcomes
  - The **net gain from a decision** can be identified and used to consider whether an investment is worthwhile
- Using decision trees provides **several key advantages** to the decision-making process:
  - Constructing a decision tree diagram may **reveal options** that haven't previously been considered
  - Managers are forced to **consider the risks** associated with their choice, ahead of implementation
  - The quantitative approach requires **deep research** to be carried out

Decision tree diagrams

- The **key elements** in a decision tree diagram are:
  - Decision points
  - Outcomes
  - Probabilities
  - Expected monetary values

A simple decision tree



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Got it

- Points where **decisions** need to be made are represented by squares
  - Square A represents the fact that a choice is required on opening a new store or expanding the website

- Points where there are different **outcomes** are represented by circles called **nodes**
  - Circles B and C represent points at which the different options have a range of outcomes — success or failure

- The **probability** or likelihood of each outcome is shown on the diagram
  - A certain outcome has a probability of 1
  - An impossible outcome has a probability of 0
    - Opening a new store has a 0.7 probability of success and a 0.3 probability of failure
    - Expanding the website has a 0.6 probability of success and a 0.4 probability of failure

- The **monetary value** of each decision is based on the expected profit or loss of the outcome
  - If opening a new store is successful, a £420,000 profit is expected
  - If opening a new store is unsuccessful, a £24,000 loss is expected
  - If expanding the website is successful, a £480,000 profit is expected
  - If expanding the website is unsuccessful, a £32,000 loss is expected

Calculating expected monetary values

- To compare the options, a business should take into account the **expected values of each decision** presented in the decision tree diagram
- To calculate the **expected monetary value** of a decision, the following formula is used

(Expected value of success × probability) + (Expected value of failure × probability)

- Using the example above, the expected value of opening a new store is

$$(£420{,}000 \times 0.7) \quad + \quad (-£24{,}000 \times 0.3)$$

$$= \quad £294{,}000 \quad + \quad -£7{,}200$$

$$= \quad £286{,}800$$

- Using the example above, the expected value of expanding the website is

$$(£480{,}000 \times 0.6) \quad + \quad (-£32{,}000 \times 0.4)$$

$$= \quad £288{,}000 \quad + \quad -£12{,}800$$

$$= \quad £275{,}200$$

- As the expected value of opening a new store is higher, at £286,800, than that of expanding the website, at

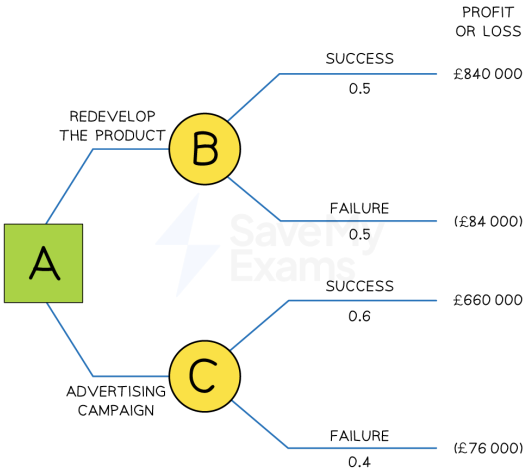
Worked Example

Caramelac is a lactose-free chocolate product manufactured by a large multinational confectionery business. In recent years, increased competition from other well-known brands has started to impact sales of the product, and managers are determined to maintain Caramelac's market share.

Market research has shown that the business has two options:

a) Redevelop the product
b) Create a new advertising campaign

The expected outcomes and the probabilities of success and failure are shown in the decision tree below.



Calculate the expected values of each option and decide, on financial grounds, which option Caramelac's brand managers should choose.

[6]

Step 1: Calculate the expected value of redeveloping the product

$$= (£840{,}000 \times 0.5) + (-£84{,}000 \times 0.5) = £420{,}000$$

$$= £420{,}000 + -£42{,}000$$ [1]

$$= £378{,}000$$

Step 2: Calculate the expected value of the advertising campaign

$$= (£660{,}000\ x\ 0.6) + (-£76{,}000\ x\ 0.4) = £396{,}000$$

$$= £396{,}000 + -£30{,}400$$ [1]

$$= £365{,}600$$

Step 3: Interpret the outcomes and make a decision

the option to redevelop the product [1].

Worked Example

- In some cases, the decision tree diagram **provides expected revenues rather than profit or loss** for the range of outcomes
- In these diagrams, **the costs related to each outcome are also provided**
- To calculate the expected value of each outcome, **costs must be deducted from expected revenues**

- To calculate the **expected monetary value** of a decision where revenues and costs are included in the diagram:

(Expected value of success × probability) + (Expected value of failure × probability) - cost

- The expected value of launching a new product is

$= (£520{,}000 \times 0.6) + (-£54{,}000 \times 0.4) - £280{,}000 = £312,$

$= £312{,}000 + -£21{,}600 - £280{,}000 = £290{,}400$

$= £290{,}400 - £280{,}000 = £10{,}400$

- The expected value of improving the existing product is

$= (£225{,}000 \times 0.9) + (-£22{,}000 \times 0.1) - £190{,}000 = £202{,}5$

$= £202{,}500 + -£2{,}200 - £190{,}000 = £200{,}300$

$= £200{,}300 - £190{,}000 = £10{,}300$

- As the expected value of launching a new product is marginally higher, at £10,400, than that of improving the existing product, at £10,300, the business should choose the option to launch a new product
- In this case, the decision tree has demonstrated that there is **little difference between the two options**, and

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Expected values are not the same thing as profit or revenues generated by one of the choices. In the above example, launching a new product is expected to either generate a positive revenue figure of £520,000 or generate a negative revenue figure of £54,000. It is never forecast that a revenue figure of £200,300 will be achieved. This is purely a figure used in making the choice between this option and the alternative and does not represent the actual amount of revenue that is expected to be achieved.

Limitations of using decision trees

- Constructing decision trees that can support effective decision-making **requires skill** to avoid bias and takes a **significant amount of time to gather reliable data**
- A decision tree is **constructed using estimates** that rarely take full account of **external factors** and **cannot include all possible eventualities**
- **Qualitative elements**, such as human resource impacts, are not considered, which may **affect the probability of success** of a decision
- The **time lag** between the construction of a decision tree diagram and the implementation of the decision is likely to further affect the **reliability of the expected values**

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